



What Does Drama Sound Like?

Sound Analysis of GerDraCor at DraCor Summit 2025

FU Berlin, 03.09.2025

Janis Pagel, Svenja Guhr, Nils Reiter

Fictional Soundscapes (Guhr 2025)

“[T]he ensemble of sounds represented in a literary text that provides information about (1) the acoustic setting of the fictional world with focus on (2) the sounds produced by the characters and their ambiance in the course of the fiction.”

Character Sounds and Ambient Sounds

Ex. 14 <sp who=“laszlo”> <speaker>Laszlo.</speaker>

<p>... jagt ... <ambient_sound>und ratter</ambient_sound> ... <ambient_sound>und knatter</ambient_sound> und wirbelt ... <stage><character_sound>Lacht auf</character_sound> und wirft die Pfeife in die Ecke.</stage></p> </sp> (*Die Haidebraut*, Stramm)

Sound Analysis of German Plays

4 types of sound indications found in GerDraCor:

Ex. 1 “Darum ich euch sehr fleißig **bitt**”

“I kindly **request** that you do so”. (*Eulenspiegel*, Sachs)

Ex. 2 “Wie du jetzt selber hast **gered’t**”

“As you have **stated** yourself”. (*Eulenspiegel*, Sachs)

Ex. 3 “<stage>Man hört noch einen **Schuß**.</stage>”

“<stage>Another **shot** is heard</stage>”. (*Die Räuber*, Schiller)

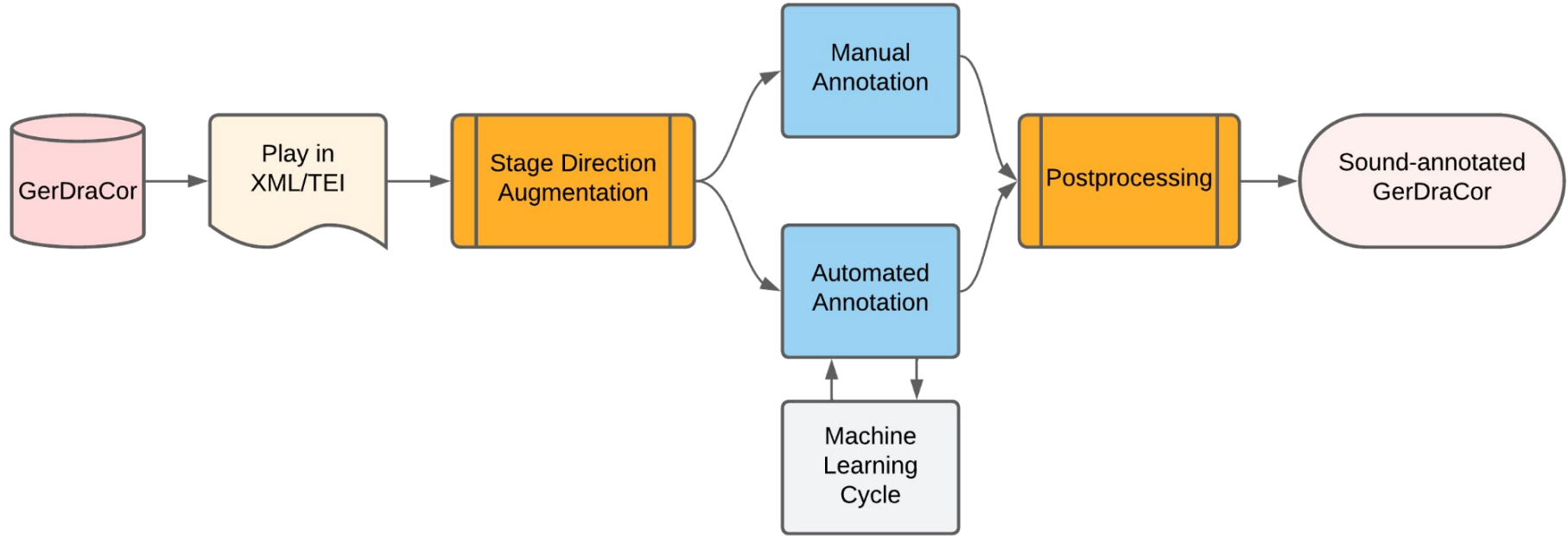
Ex. 4 “ORSINA <stage>**heftig**.</stage> Nicht gelesen?”

“ORSINA <stage>**fiercely**</stage> Didn’t read?”. (*Emilia Galotti*, Lessing)

Research Questions

- 1) How does sound in drama compare to sound in prose?
- 2) How can detecting sound be operationalized in dramatic texts?
- 3) How is sound used in plays across different (sub-)genres and centuries?

Method



Augmentation of the Stage Directions

Ex. 5 <sp who="nathanael_jettel"> <speaker>Nathanael Jettel,</speaker>
<stage>sehr ruhig.</stage><p>Könnten Sie mir nicht sagen, Direktor, wer Ihnen in Gottes Namen auf die Krawatte
getreten hat?</p> </sp> (*Die Ratten*, Hauptmann)

Ex. 6 <sp who="nathanael_jettel"> <speaker>Nathanael Jettel,</speaker>
<stage>**sagt** sehr ruhig.</stage> <p>Könnten Sie mir nicht sagen, Direktor, wer Ihnen in Gottes Namen auf die
Krawatte getreten hat?</p> </sp> (*Die Ratten*, Hauptmann)

Training Set for Finetuning German BERT (Chan et al. 2020)

	Author (Publication Year)	Title
Training Set	Ayrer (1605)	<i>Fassnachtspiel – Wie einem Weib jhr eygener Mann</i>
	Neuber (1736)	<i>Die beschützte Schauspielkunst</i>
	Lessing (1772)	<i>Emilia Galotti</i>
	Schiller (1781)	<i>Die Räuber</i>
	Günderode (1805)	<i>Udohla</i>
	Chezy (1823)	<i>Der neue Narziss</i>
	Ebner-Eschenbach (1862)	<i>Die Veilchen</i>
	Dohm (1878)	<i>Ein Schuß in's Schwarze</i>
	Dovsky (1915)	<i>Mona Lisa</i>
	Borchert (1947)	<i>Draußen vor der Tür</i>
Test Set	Sachs (1557)	<i>Eulenspiegel mit dem blauen Hosentuch</i>
	Wedekind (1891)	<i>Frühlings Erwachen</i>

Table 1

Set of 12 manually annotated plays distributed to separate training and test sets.

Automated Sound Annotation - 5 Experiments

TEI-enriching NEISS NTEE software (by Lemke, Sperfeld, Zöllner)

Training set	65 prose	65 prose + 5 drama	5 prose + 5 drama	5 drama	10 drama
Tokens	736,970	835,750	237,686	98,780	151,314
Sound events	10,763	11,876	2,896	1,113	1,761
SequEval-F1 ⁴	0.01 / 0.15	0.20 / 0.20	0.01 / 0.23	0.00 / 0.33	0.43 / 0.53
Precision	0.10 / 0.23	0.39 / 0.29	0.07 / 0.32	0.04 / 0.37	0.36 / 0.54
Recall	0.07 / 0.32	0.26 / 0.34	0.07 / 0.36	0.04 / 0.46	0.61 / 0.52
F1-Score	0.09 / 0.27	0.31 / 0.32	0.07 / 0.34	0.04 / 0.41	0.45 / 0.53
E-F1-Score	0.02 / 0.24	0.21 / 0.28	0.02 / 0.39	0.00 / 0.48	0.54 / 0.60

Table 3

Evaluation scores on the test texts: 1) Sachs' *Eulenspiegel* and 2) Wedekind's *Frühlings Erwachen*. The first evaluation score is always Sachs, the second Wedekind.

What Can We Do with Sound Annotations?

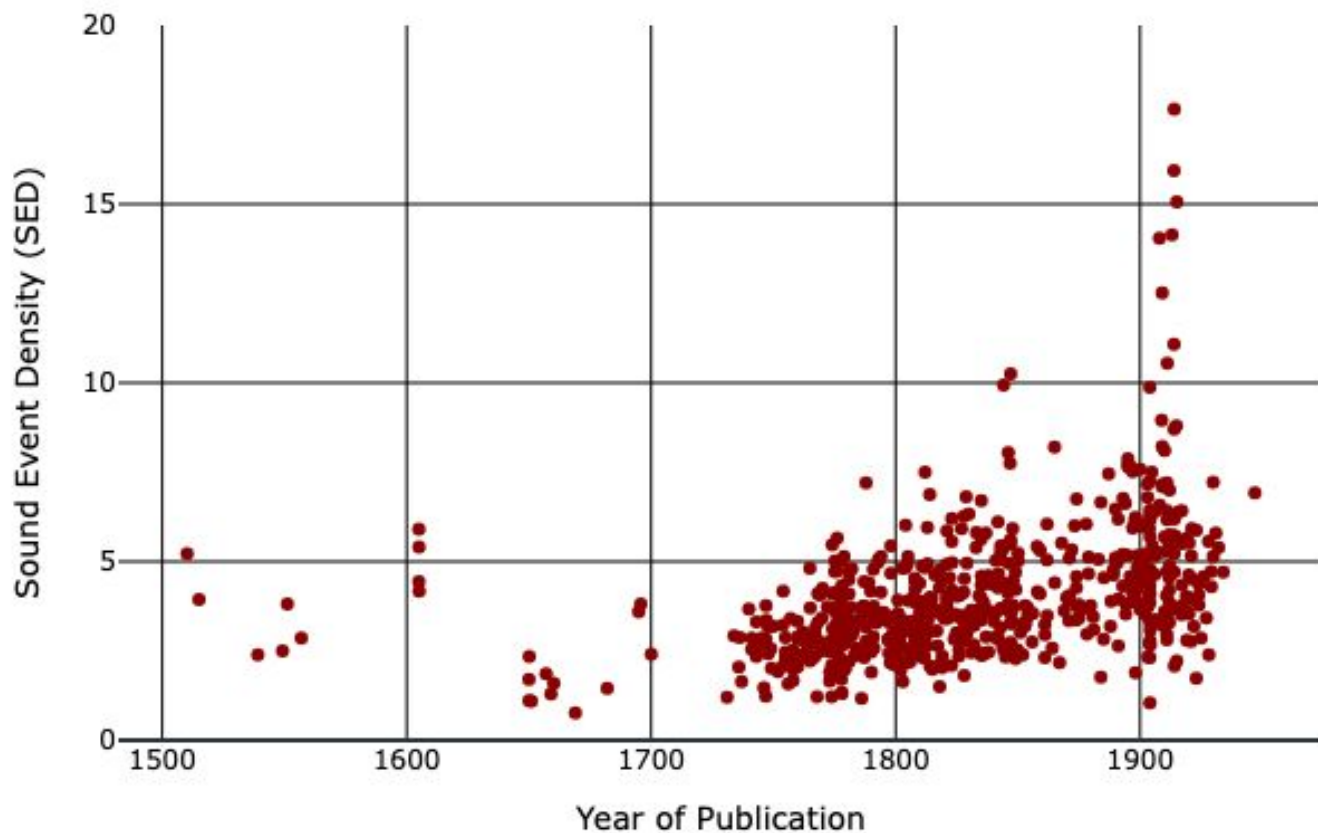
Sound Event Density Calculation (SED) developed by Guhr (2025)

T_{se} : absolute count of annotated sound events

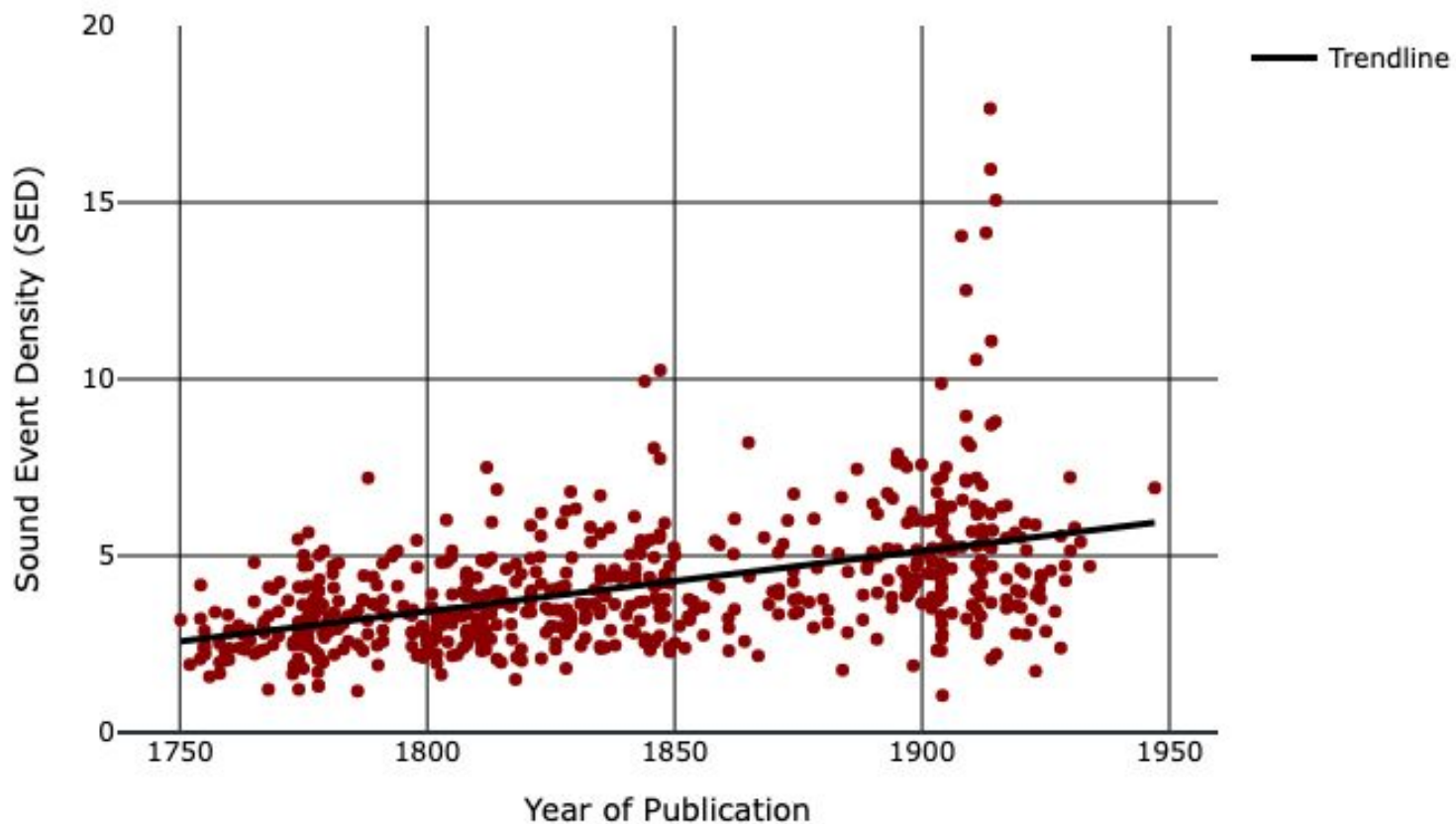
T_{pe} : number of potential events in a text

$$SE_D = \frac{T_{se}}{T_{pe}} \times 100$$

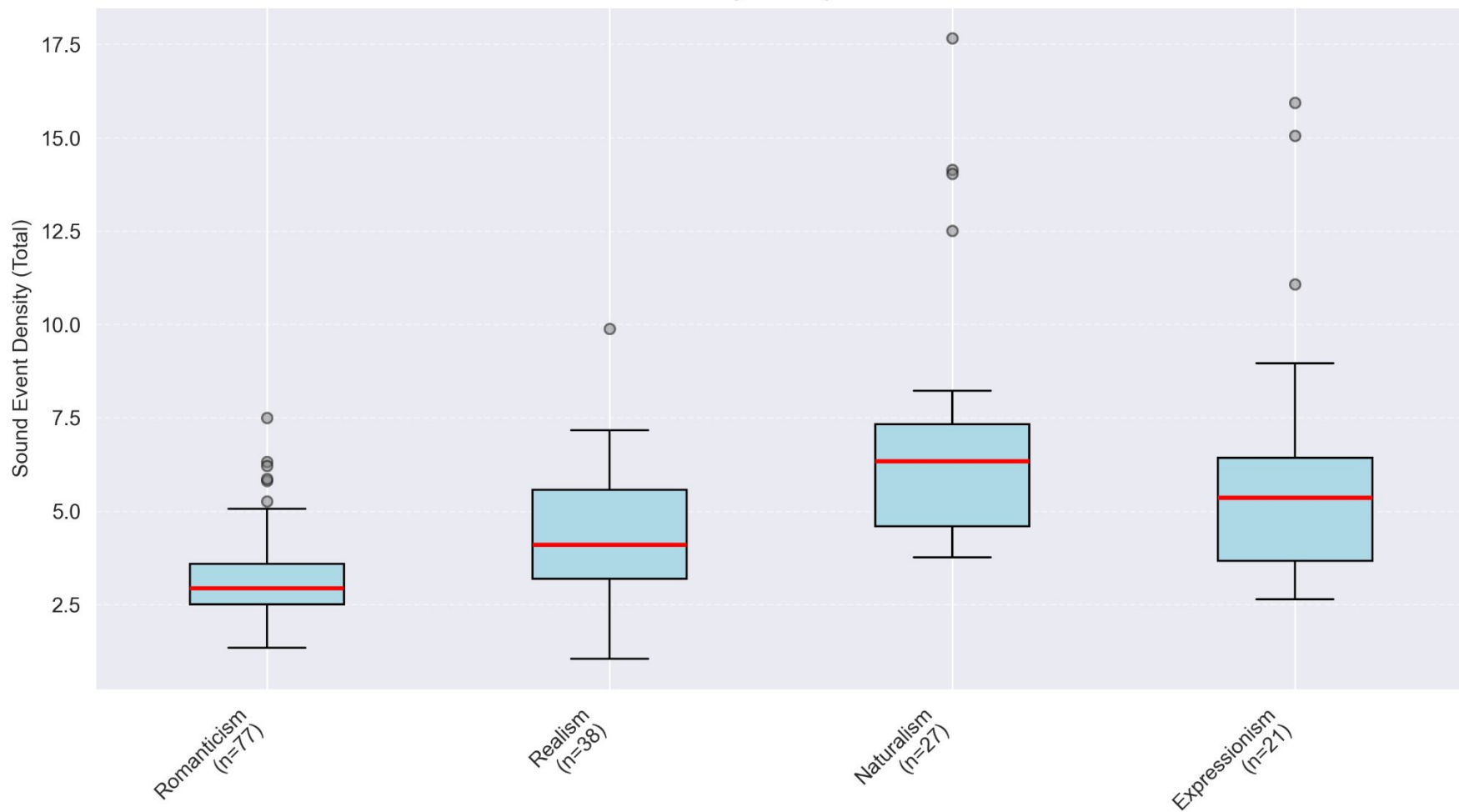
Sound Event Density per Text by Year (All Genres, n=649)



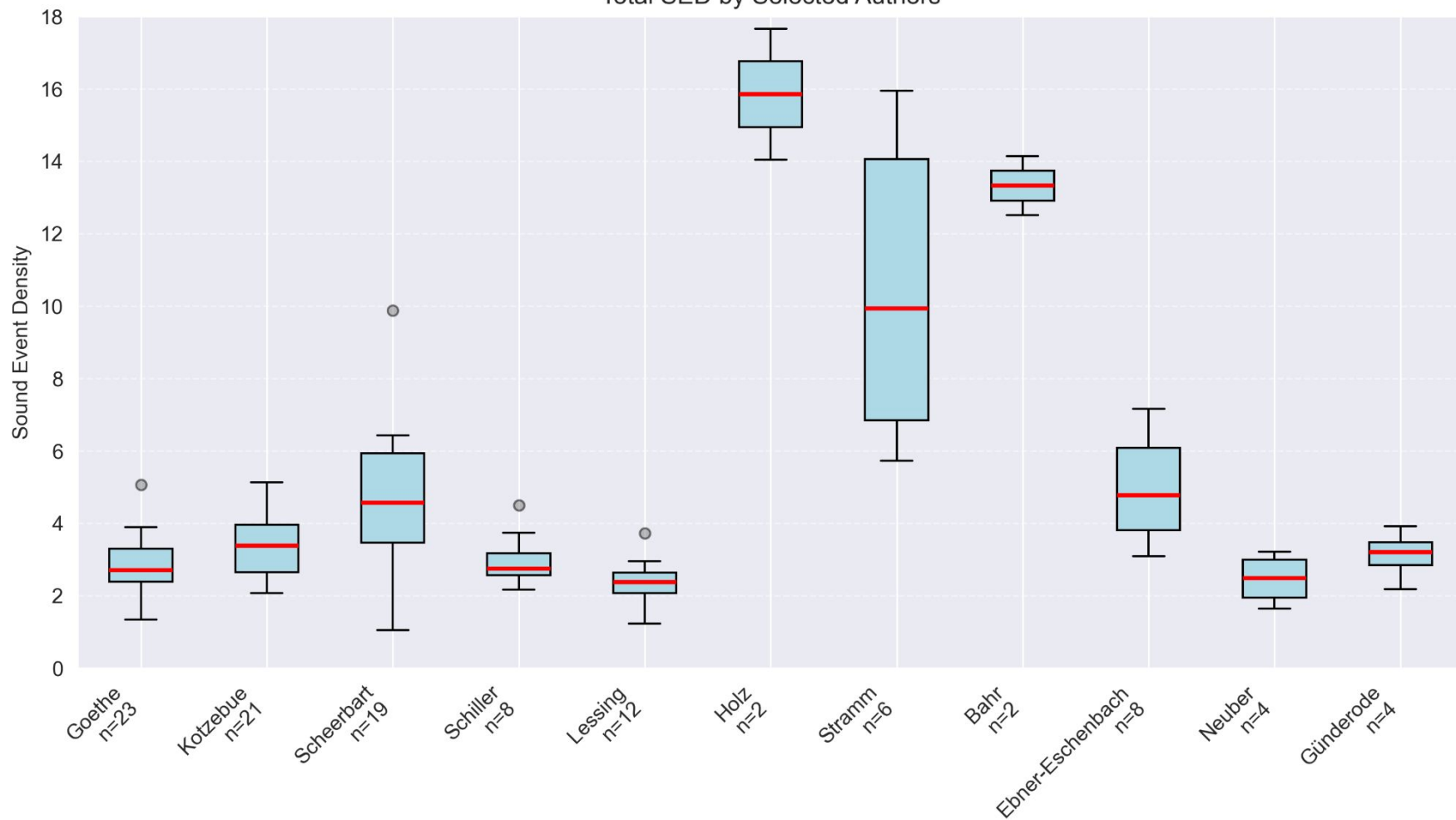
Sound Event Density per Text by Year (1750-1950, n=604)



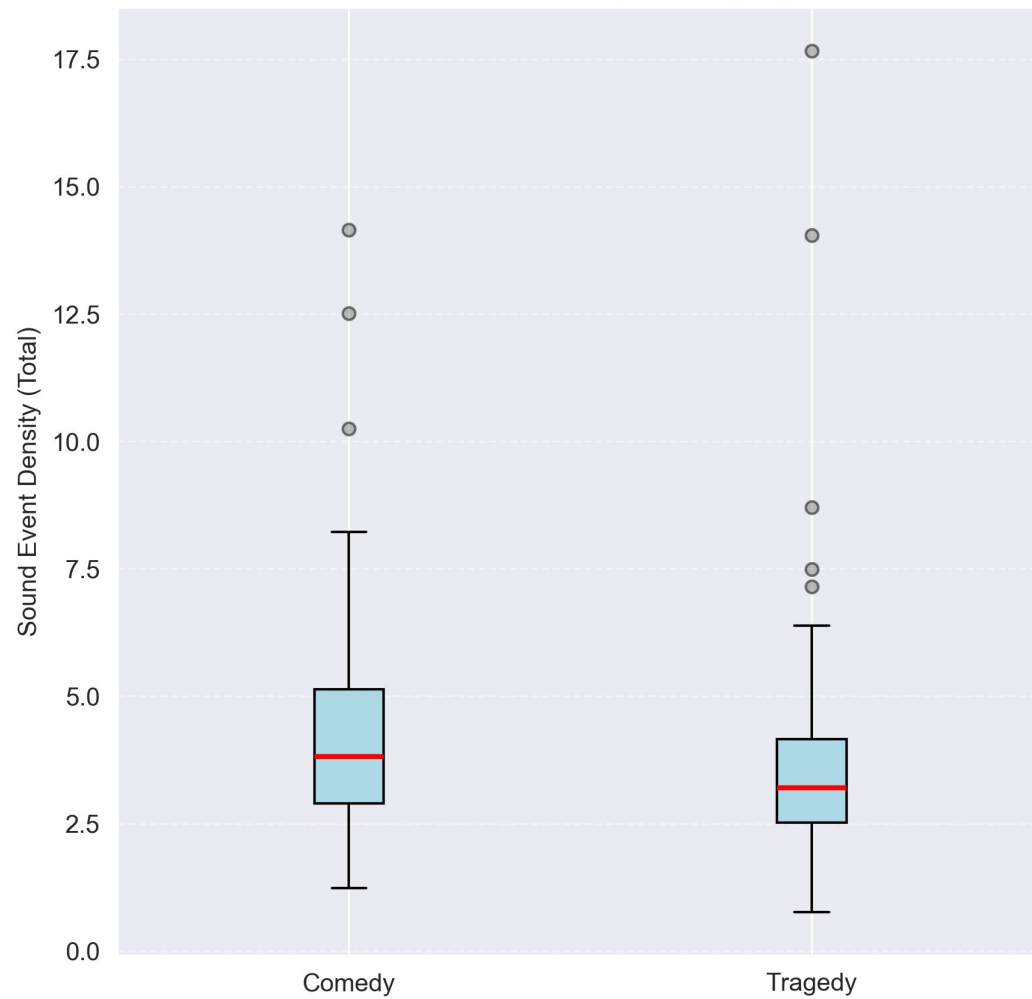
Total SED by Literary Movement



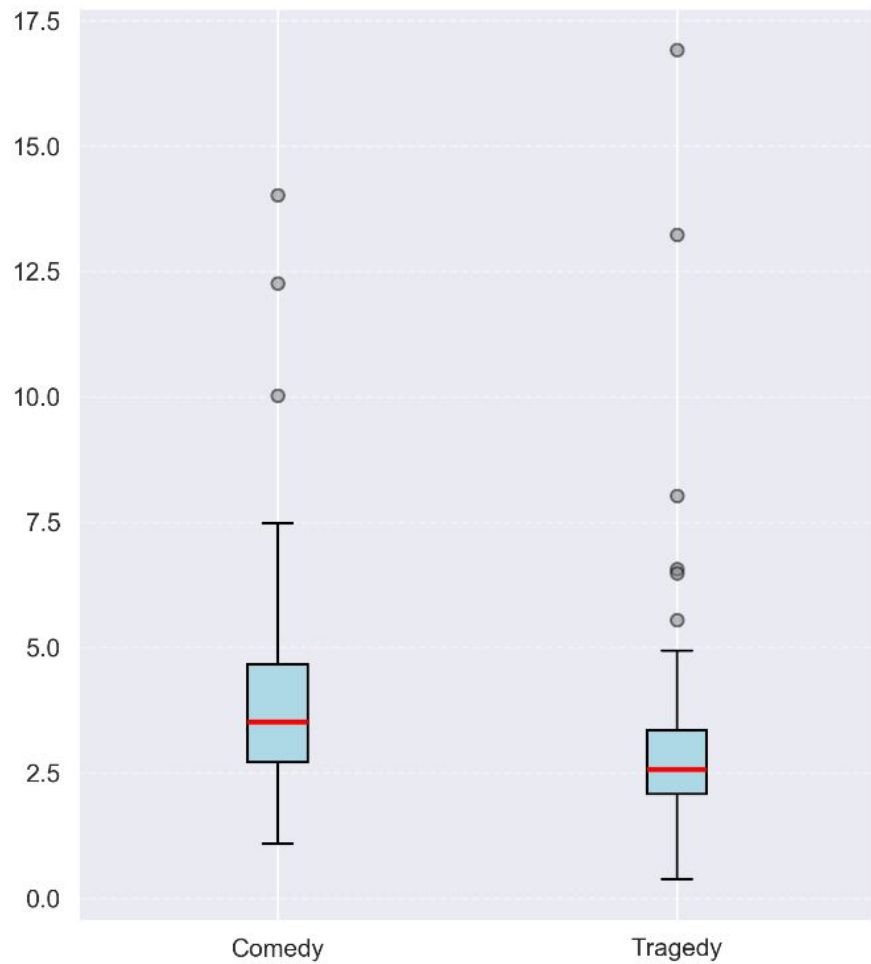
Total SED by Selected Authors



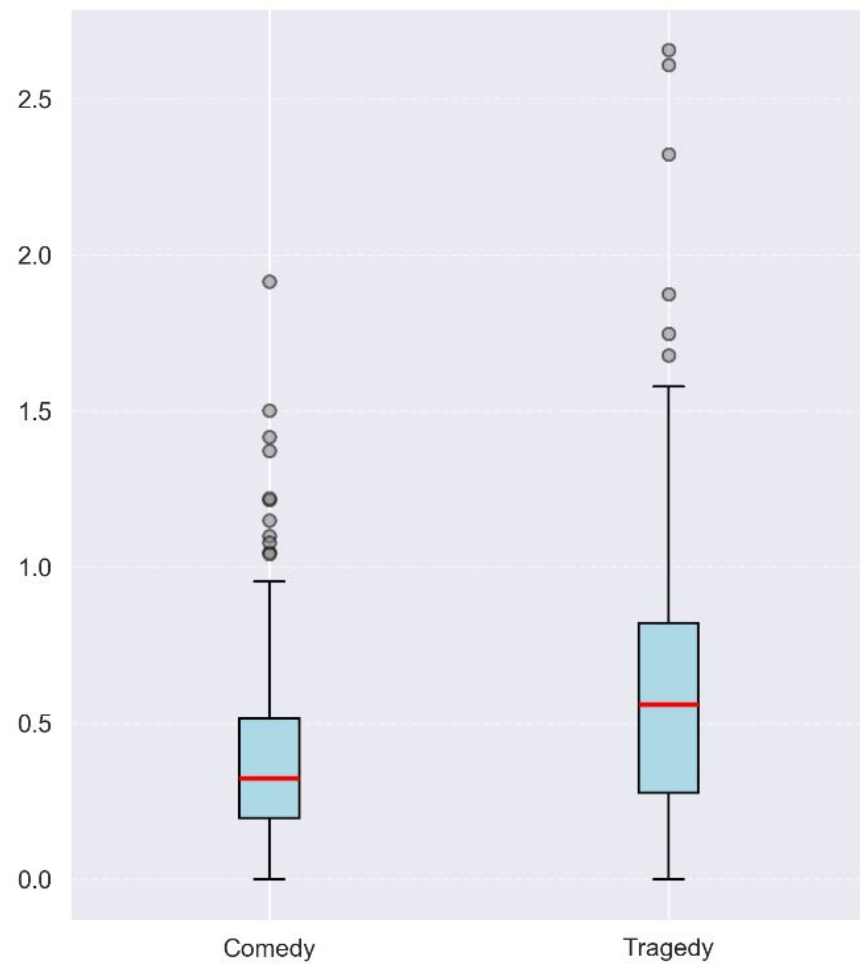
Total SED: Comedy vs. Tragedy



Character SED

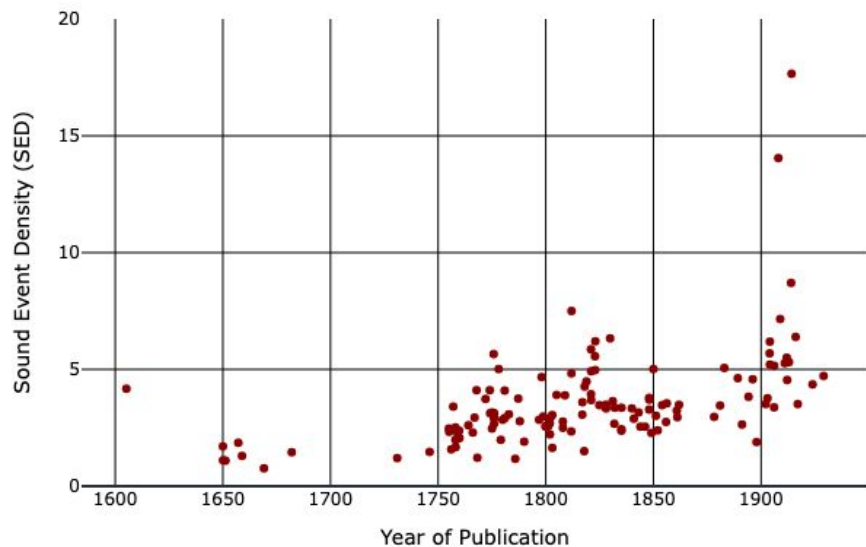


Ambient SED

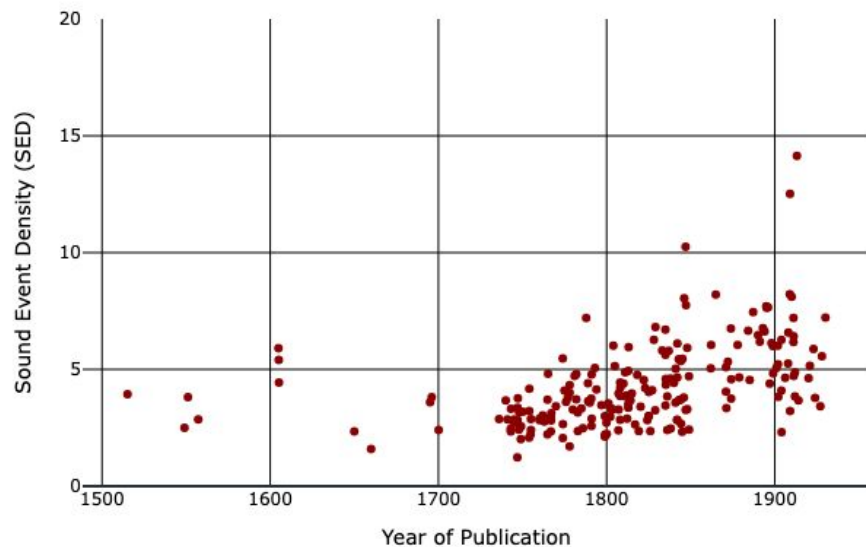


SED over the Course of GerDraCor by Subgenre

Sound Event Density per Text by Year (Tragedy, n=132)



Sound Event Density per Text by Year (Comedy, n=211)



Conclusion and Outlook

- Re-using prosa annotations for model training does not help
 - Sound representation in drama offers new challenges for sound annotation
 - Heuristic worked out well - next step is to try the automation without stage direction augmentation → more manually sound-annotated plays needed
 - Over the course of GerDraCor sound representation increases significantly
 - Tragedies and comedies work differently sound-wise
 - Current approach focuses on frequency analysis based on SED calculations
- New content-oriented research questions arise, e.g.:
- What kind of sounds are represented?
 - Are there sound-groups beyond character and ambient sounds?
 - Do characters of different gender produce different sounds?
- Add loudness levels for comparative analysis



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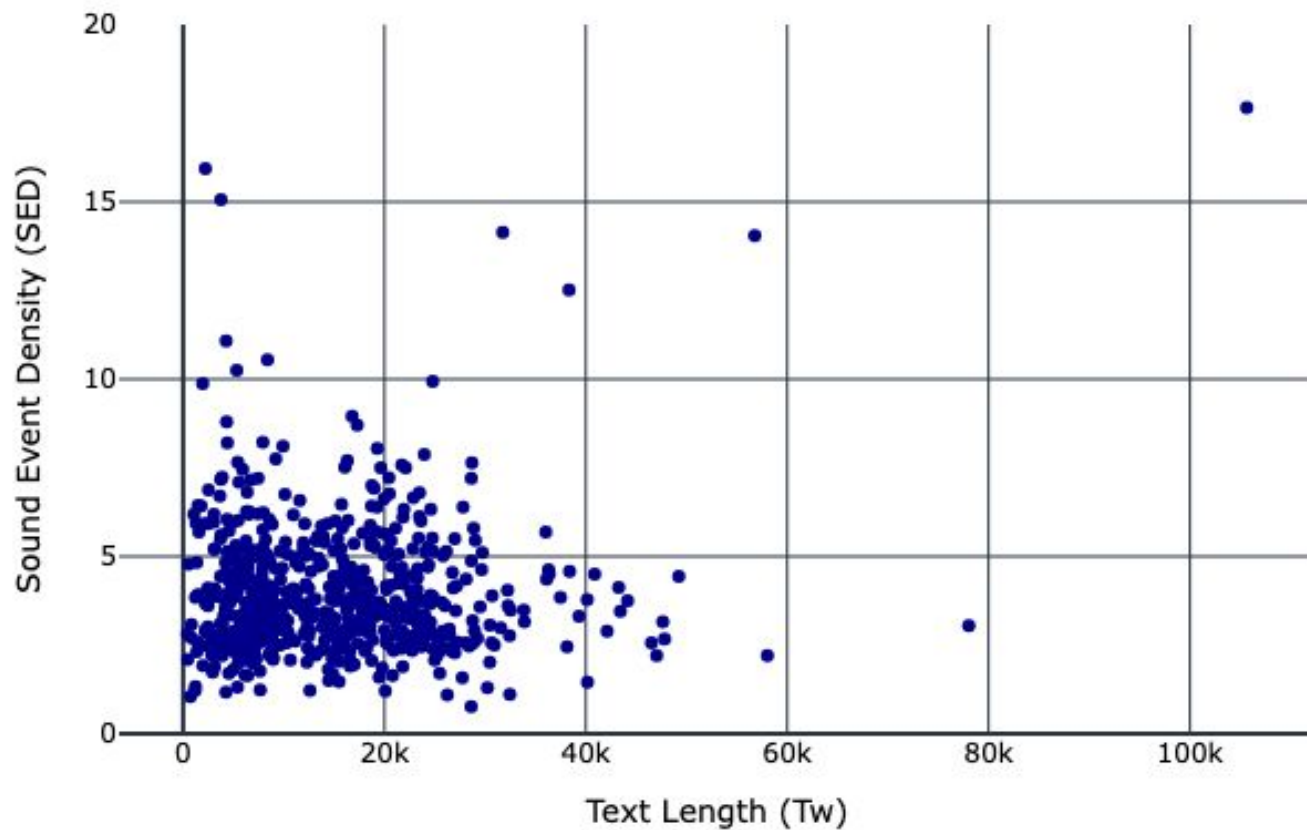
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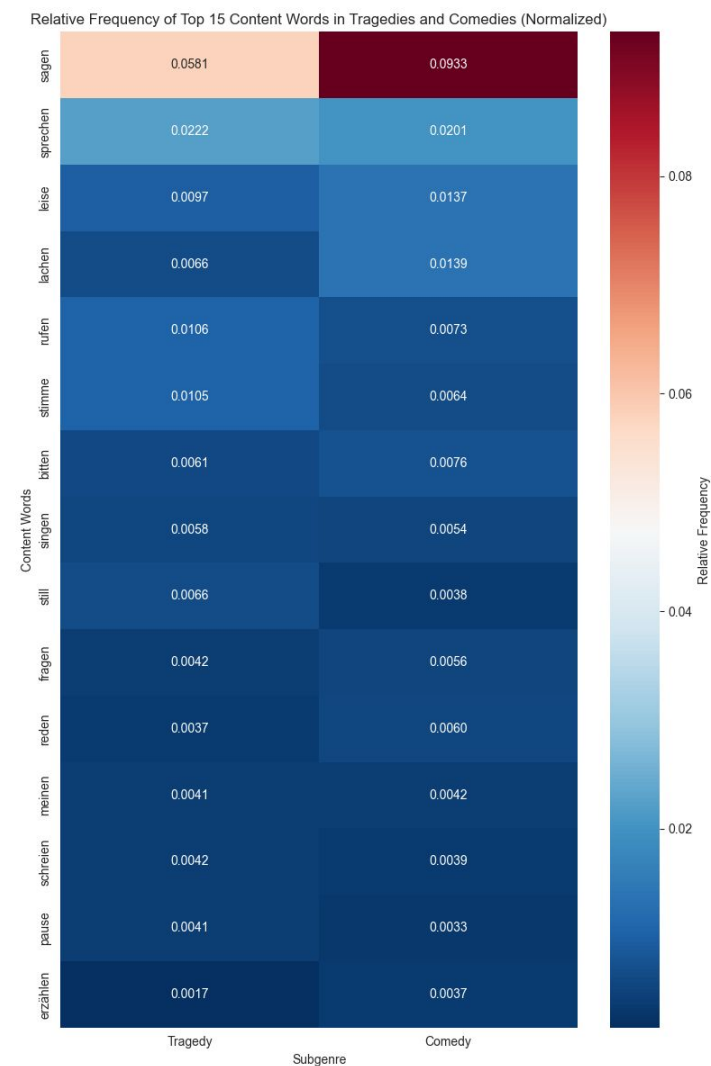
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Appendix

Sound Event Density vs. Text Length



Most Frequent Words per Subgenre



Interesting Sample Passages

- reported speech:

<p><character_sound>Ich sage dir alles</character_sound>. – Ich habe es teils aus Büchern, teils aus Illustrationen, teils aus Beobachtungen in der Natur. Du wirst überrascht sein; ich wurde seinerzeit Atheist. <character_sound>Ich habe es auch Georg Zirschnitz gesagt</character_sound>! Georg Zirschnitz wollte es Hänschen Rilow sagen, aber Hänschen Rilow hatte als Kind schon alles von seiner Gouvernante erfahren.</p> (Wedekind, *Frühlings Erwachen*)

- reported ambient sounds:

<p>[...] Du ziehst bei mir ein wie die angestammte Herrin in ihr verödetes Schloß. Tor und Türen öffnen sich von unsichtbarer Hand, <ambient_sound>während der Springquell unten im Parke fröhlich zu plätschern beginnt</ambient_sound> ...</p> (Wedekind, *Frühlings Erwachen*)

Interesting Sample Passages

- sound descriptions in stage directions

<sp who="#frau_bergmann">

<speaker>FRAU BERGMANN</speaker>

<stage><character_sound>ekstatisch.</character_sound></stage>

<p>– Aber es geht ja nicht, Kind! – Ich kann es ja nicht verantworten. – Ich verdiene ja, daß man mich ins Gefängnis setzt – daß man dich von mir nimmt ...</p> </sp> (Wedekind, *Frühlings Erwachen*)

- ambiguous descriptions in stage directions – greetings spoken or as gestures?

<stage>Melchior geht vorüber und grüßt.</stage> (Wedekind, *Frühlings Erwachen*)